

Simulation Use Limitation Module (SUM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Parent Standard: Simulation Evidence & Reliance Boundary Standard (SERBS)

Module: SUM — Simulation Use Limitation Module

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Governed Space: Simulation Evidence Use Limitation

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AIG® · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodology Intellectual Property Protection

Canonical Language: English

Canonical Definition

Simulation Use Limitation Module (SUM) is the fourth Core Module of the Simulation Evidence & Reliance Boundary Standard (SERBS) within SIMULOS® — Simulation Governance Architecture.

SUM governs the explicit restrictions, exclusions, conditions, caveats, unsupported interpretations, prohibited uses, and dependency conditions that must remain attached to simulation-derived evidence when that evidence is interpreted, communicated, transferred, or used.

The Module translates the governed reliance boundary established by SRBM — Simulation Reliance Boundary Module into explicit controls over what simulation evidence may not legitimately be used to claim, infer, justify, demonstrate, predict, authorize, or support.

SUM does not determine whether simulation evidence qualifies as evidence, how much evidential weight it carries, or whether the complete validation basis is sufficient.

Its task is narrower:

to ensure that the limitations governing simulation-derived evidence remain explicit, persistent, interpretable, transferable, and inseparable from

downstream use of that evidence.

Core Question

What must simulation-derived evidence not be used to claim, infer, justify, or support?

Architectural Position

SUM is the fourth module within SERBS.

It follows:

SEQM — Simulation Evidence Qualification Module

→ establishes what qualifies as Simulation Evidence;

SEWM — Simulation Evidence Weight Module

→ establishes how much simulation-specific evidential significance the evidence may carry;

SRBM — Simulation Reliance Boundary Module

→ establishes how far legitimate reliance may extend.

SUM then converts those boundaries into explicit use controls.

It precedes:

SVIM — Simulation Validation Interface Module

which governs transfer of simulation-derived evidence into broader validation governance under VALIDOS®.

The sequence is therefore:

Qualification → Weight → Reliance Boundary → Use Limitation → Validation Interface.

Internal Flow Position

Weighted Simulation Evidence Governed Simulation Reliance Boundary → Intended Use Identification → Limitation Source Identification → Use Condition Definition → Restriction Definition → Exclusion Definition → Unsupported Interpretation Identification → Prohibited Use Definition → Limitation Persistence Requirements → Use Limitation State → Governed Simulation Use Limitation Record → SVIM — Simulation Validation Interface Module

Operational Space

SUM governs the space between:

where legitimate simulation reliance ends

and

how that boundary must constrain actual evidence use.

This operational space includes: intended evidence use; permitted-use conditions; conditional use; restricted use; excluded use; unsupported interpretation; prohibited inference; prohibited generalization; prohibited extrapolation; decision-class limitation; configuration limitation; scenario limitation; parameter limitation; environmental limitation; temporal limitation; uncertainty limitation; correlation limitation; dependency limitation; communication limitation; transfer limitation; limitation persistence; limitation review; limitation withdrawal following material change.

Operational Scope

SUM applies whenever simulation-derived evidence is: interpreted; summarized; aggregated; communicated; transferred; presented; cited; compared; used in engineering; used in scientific analysis; used in safety arguments; used in mission planning; used in design decisions; used in system qualification; used in regulatory submissions; used in validation processes; used in operational preparation; used in deployment decisions; used in AI or autonomous-system evaluation; used in consequential organizational decisions.

It applies to evidence derived from: individual simulation runs; repeated executions; Monte Carlo studies; scenario portfolios; simulation campaigns; engineering simulations; scientific simulations; safety simulations; mission simulations; digital twins; physical simulations; hybrid simulations; AI simulations; autonomous-system simulations; future simulation technologies.

SUM remains technology-neutral. Reliance Boundary ≠ Use Limitation

SUM enforces the distinction:

Reliance Boundary ≠ Use Limitation

SRBM determines:

where legitimate reliance stops.

SUM determines:

what restrictions must govern use because that boundary exists.

For example, SRBM may establish that evidence applies only to:

Object Version V4 + Scenario Family S3 + Parameter Range P2 + Environment E1.

SUM may translate this into explicit controls:

Do not apply this evidence to Object Version V5.

Do not generalize this evidence beyond Scenario Family S3.

Do not use this evidence to support performance outside Parameter Range P2.

Do not represent this evidence as demonstrating performance in environments materially different from E1.

SRBM defines the boundary.

SUM operationalizes it.

Use Limitation Principle

A Simulation Use Limitation is:

an explicit governance condition constraining how simulation-derived evidence may be interpreted, communicated, transferred, generalized, or relied upon.

A limitation may arise from: simulation purpose; simulation scope; exclusions; reality representation; abstraction; fidelity; assumptions; model applicability; environmental configuration; input basis; scenario coverage; execution conditions; output classification; inference basis; simulation-to-reality correlation; uncertainty; evidence weight; reliance boundary; temporal relevance.

Use limitations must remain linked to their source.

Limitations Are Part of the Evidence

SUM establishes a central principle:

A material limitation is part of the governed evidential meaning, not optional commentary attached to it.

Removing a material limitation while preserving the evidence can materially change the meaning of the evidence.

Therefore:

Evidence – Material Limitation ≠ Same Evidence Meaning.

Where a limitation materially determines legitimate interpretation, the limitation must travel with the evidence.

Intended Use Identification

SUM requires the intended use of simulation evidence to be explicit.

An Intended Use may include: exploration; comparison; screening; engineering design; optimization; hypothesis support; safety analysis; scenario planning; mission planning; training; system qualification; validation input; operational preparation; deployment support; regulatory evidence; predictive analysis.

The same evidence may have different limitations for different intended uses.

Permitted Use Conditions

SUM may establish conditions under which evidence remains legitimately usable.

Examples include:

Permitted only for comparative analysis.

Permitted only for the governed Simulation Object configuration.

Permitted only within represented environmental conditions.

Permitted only within Parameter Range P.

Permitted only while Assumption Set A remains applicable.

Permitted only as one input to a broader validation process.

Conditional use must remain distinguishable from unrestricted use. Restriction

A restriction narrows how evidence may be used without completely prohibiting its use.

Restrictions may concern: purpose; object; configuration; scenario; environment; parameter range; decision type; temporal period; inference class; communication context.

For example:

Evidence may support comparative design selection but must not be used as independent evidence of operational safety. Exclusion

An exclusion identifies a condition or domain that the evidence does not cover.

Examples include: weather conditions not represented; hardware versions not simulated; excluded failure modes; populations not modeled; untested terrain; unrepresented physical phenomena; mission phases outside simulation scope; operating conditions outside parameter limits.

An exclusion must not disappear merely because downstream users are unaware of it.

Unsupported Interpretation

An Unsupported Interpretation occurs where the evidence is described as meaning more than its governed basis supports.

Examples include:

Simulation demonstrated safe behavior

when the evidence only establishes:

no unsafe behavior was observed within the simulated scenario set.

Or:

The system performs reliably in reality

when the evidence establishes only:

high simulated performance within represented conditions.

SUM requires unsupported interpretations to be identifiable and preventable.
Observed Absence $\neq$ Demonstrated Impossibility

Simulation campaigns frequently produce no instance of a particular failure.

SUM establishes:

Failure Not Observed $\neq$ Failure Impossible

The absence of an event within a simulated scenario space may provide evidence concerning that tested space.

It does not automatically establish impossibility outside that space.

This distinction is especially important in: rare-event simulation; safety analysis; autonomous systems; mission simulation; infrastructure failure analysis; high-consequence engineering.

Simulation Success $\neq$ Operational Safety

A simulation campaign may demonstrate extensive successful simulated behavior.

That does not independently establish operational safety.

Therefore:

Simulation Success $\neq$ Operational Safety Determination.

SUM must prevent simulation success metrics from being communicated as broader safety determinations where the evidence basis does not support that interpretation. Scenario Success $\neq$ Universal Performance

Successful performance across all executed scenarios means:

successful performance across those governed executed scenarios.

It does not mean:

successful performance across every possible condition.

SUM therefore preserves:

Tested Scenario Space $\neq$ Total Possible Reality Space. Comparative Evidence $\neq$ Absolute Evidence

Evidence demonstrating that:

Configuration A outperformed Configuration

B

does not necessarily establish:

Configuration A meets an absolute operational performance requirement.

SUM requires comparative claims and absolute claims to remain distinct.
Simulation Prediction $\neq$ Guaranteed Outcome

Where simulation evidence supports prediction, SUM requires the distinction:

Predicted Outcome $\neq$ Guaranteed Outcome.

Predictive simulation evidence may legitimately inform planning and decision-making.

It must not acquire deterministic meaning merely because the simulation produces precise numerical outputs.

Precision Limitation

Highly precise simulation outputs may create false confidence.

SUM therefore establishes:

Numerical Precision $\neq$ Evidential Certainty.

An output of:

0.003721 failure probability

does not necessarily mean the real-world probability is known to six significant digits.

The precision communicated must remain compatible with: model uncertainty; input uncertainty; scenario uncertainty; representation limits; correlation uncertainty; inference uncertainty.

Extrapolation Limitation

Where evidence is extrapolated beyond directly simulated conditions, the extrapolation must remain explicitly identified.

SUM may require statements such as:

This conclusion includes extrapolation beyond the directly simulated parameter range.

or:

This evidence does not independently support performance beyond the simulated environmental domain.

Unidentified extrapolation is a use-governance failure.

Decision-Class Limitation

Simulation evidence may legitimately inform one decision class while remaining insufficient for another.

For example:

Exploratory Design Decision

may require less evidential support than:

Safety-Critical Deployment Decision.

SUM may therefore assign different use limitations based upon the intended decision class.

It does not define organizational decision authority.

Configuration Limitation

Evidence associated with one Simulation Object configuration must retain limitations concerning materially different configurations.

Changes may include: hardware; software; architecture; sensor suite; actuator system; material; payload; model configuration; operating mode.

Evidence transfer across configurations requires a governed basis.

Scenario Limitation

SUM requires scenario limitations to remain visible.

Evidence may be limited to: expected scenarios; edge cases; stress conditions; specified scenario families; defined uncertainty ranges; specific failure combinations.

A scenario limitation should identify both:

what was represented

and, where material:

what was not represented.

Environmental Limitation

Evidence may be valid only under specified represented environments.

Limitations may concern: weather; terrain; lighting; atmosphere; gravity; thermal conditions; traffic; infrastructure; communication availability; human interaction; electromagnetic conditions.

Environmental limitations are especially important where simulation-to-reality transfer is material.

Temporal Limitation

Simulation evidence may have a temporal use boundary.

A limitation may require review when: the Simulation Object changes; the

operational environment changes; new reality data becomes available; model versions change; assumptions become obsolete; new failure modes emerge; input distributions shift; correlation deteriorates.

Evidence should not retain permanent unrestricted use merely because it was once appropriately qualified.

Uncertainty Limitation

Material uncertainty must constrain interpretation where relevant.

SUM may require: confidence qualification; uncertainty ranges; conditional language; restricted extrapolation; additional evidence; prohibition of deterministic claims.

Uncertainty is not a reason to discard useful evidence.

It is a reason to govern how strongly that evidence is expressed and used.

Dependency Limitation

Where evidence objects share material dependencies, downstream users must not interpret their quantity as equivalent independent confirmation.

SUM may therefore require disclosure that: runs share the same model; evidence shares the same assumptions; outputs share the same input source; models share a common representation; scenario sets share common omissions; evidence shares one correlation basis.

Multiple dependent evidence objects must not be presented as independent confirmation.

Aggregation Limitation

Aggregated simulation metrics must not erase material information.

An aggregate success rate may conceal: rare failures; scenario-specific failures; environmental clusters; configuration-specific weaknesses; anomalous executions; distribution tails; common-mode failures.

SUM may therefore require that material underlying limitations remain attached to aggregate evidence.

Communication Limitation

Simulation evidence can be methodologically distorted through communication even when the underlying evidence remains unchanged.

SUM therefore applies to: reports; dashboards; executive summaries; regulatory submissions; presentations; scientific communication; engineering documentation; automated AI-generated summaries; machine-readable evidence transfers.

A summary must not remove limitations necessary to interpret the evidence correctly.

Human-Readable and Machine-Readable Limitations

Because simulation evidence may move through automated systems, limitations should be capable of being represented in both:

human-readable form

and

machine-readable form.

Machine-readable limitation fields may include: permitted use; restricted use; prohibited use; applicable object; applicable version; applicable configuration; scenario domain; parameter domain; environmental domain; temporal validity; uncertainty condition; correlation condition; review trigger.

This allows limitations to survive automated evidence pipelines.

Limitation Persistence Principle

SUM establishes:

A limitation must travel at least as far as the evidence it limits.

If evidence is: copied; exported; aggregated; summarized; transferred; transformed; entered into another system; entered into validation; reused in another analysis,

material limitations must remain discoverable and linked.

Limitation Detachment

Limitation Detachment occurs when evidence continues downstream while its governing limitations are lost, hidden, omitted, weakened, or separated.

Examples include: copying a simulation metric without its scenario boundary; presenting a success percentage without excluded failures; transferring evidence without uncertainty; using a graph without its parameter range; citing a simulation conclusion after the Simulation Object configuration changed; removing caveats from an executive summary.

SUM treats material limitation detachment as a simulation governance failure.

Limitation Conflict

Different evidence objects may carry conflicting use limitations.

One evidence object may support a proposition under one condition while another constrains it under another.

SUM requires such conflicts to remain visible rather than collapsing them into one apparently unrestricted conclusion.

Use Limitation States

SUM may assign a governed Use Limitation State.

A minimum state model may include: 1.

Undefined Limitation

Evidence exists, but material use limitations have not yet been explicitly established. 2. Conditional Use

Evidence may be used only while defined conditions remain satisfied. 3. Restricted Use

Evidence may be used only for specified purposes, contexts, objects, scenarios, or decision classes. 4. Prohibited Use

A specified use, interpretation, inference, or decision application is not supported by the governed simulation evidence. 5. Review Required

Material change or new information requires reassessment before continued use. 6. Limitation Superseded

A previous limitation has been replaced by a newly governed limitation basis.

These states govern evidence use rather than overall validation status.

Prohibited Use

A Prohibited Use is a use that exceeds the legitimate simulation-specific evidential basis sufficiently that it should not proceed without a new governed basis.

Examples may include: claiming universal behavior from bounded scenarios; claiming real-world equivalence from simulation alone; claiming operational safety solely from simulated success; applying evidence to a materially different object configuration; concealing known scenario exclusions; treating correlated simulations as independent evidence; presenting inferred results as directly observed results; using stale evidence after material reality change; treating extrapolation as direct simulation evidence; representing simulation evidence as a formal Validation Determination.

Prohibited Use ≠ Organizational Prohibition

SUM defines methodological prohibited use within simulation evidence governance.

It does not itself impose legal, regulatory, corporate, or disciplinary sanctions.

Other governance structures may use SUM outputs to establish such controls.

Limitation Review Trigger

Use limitations should be reassessed when materially relevant changes occur.

Triggers may include: Simulation Object change; configuration change; new simulation results; new scenarios; new model; new input basis; new correlation evidence; operational observations; newly discovered failure; changed intended use; changed decision class; changed environment; expired temporal applicability.

Review may result in:

limitation retained → limitation narrowed → limitation expanded → limitation replaced → additional simulation required.

SUM Boundary with SRBM

SRBM determines the maximum legitimate reliance boundary.

SUM operationalizes that boundary through explicit restrictions and prohibited uses.

Therefore:

SRBM = Where reliance stops.

SUM = What use must be constrained because it stops there.

SUM Boundary with SVIM

SUM establishes the limitations.

SVIM ensures that those limitations remain attached when evidence crosses into broader validation governance.

SUM therefore provides a critical input to SVIM.

SUM Boundary with VALIDOS®

SUM does not determine validation sufficiency.

It may establish that simulation evidence:

must not independently be used to support a Validation Determination.

VALIDOS® then determines whether the total evidence basis is sufficient.

Therefore:

Simulation Use Limitation ≠ Validation Determination.

SUM Boundary with SOGS

SOGS governs limitations associated with what simulation outputs themselves mean.

SUM governs limitations on evidential use after those outputs have become qualified Simulation Evidence.

Therefore:

Output Limitation ≠ Evidence Use Limitation.

An output limitation may become an input into SUM, but the governed spaces remain distinct.

SUM Boundary with SRCS

SRCS governs simulation-to-reality correlation.

SUM may impose use limitations based upon the correlation conditions established by SRCS.

It does not determine correlation itself.

SUM Boundary with ORA™

ORA™ governs Operational Reality.

SUM may prohibit or restrict claims about Operational Reality where simulation evidence does not support them.

It does not define Operational Reality.

SUM Boundary with INTEGROS®

SUM does not govern general evidence integrity.

INTEGROS® may govern whether evidence and limitations retain integrity through processing or transfer.

SUM governs what limitations must exist and remain associated with simulation evidence.

SUM Boundary with ADIT®

ADIT® may audit whether limitations were preserved, communicated, or followed.

SUM defines the simulation-specific limitations that should govern evidence use.

SUM Boundary with AGA™

AGA™ governs who holds responsibility and accountability for decisions involving evidence.

SUM governs what simulation evidence may legitimately be used to support.

It does not assign accountability for violating a limitation. SUM Boundary with AIG® / ASGA™

SUM may constrain claims derived from simulations of AI or autonomous systems.

It does not govern the actual deployed system's behavior, autonomy, authority,

permissions, or operational constraints.

Minimum Implementation Framework

A minimum SUM implementation should establish: 1. Evidence Input

Receive qualified, weighted, reliance-bounded Simulation Evidence. 2. Intended Use Identification

Define how the evidence is intended to be interpreted or used. 3. Limitation Source Identification

Identify the simulation conditions creating material use limitations. 4. Permitted Use Conditions

Define conditions under which use remains legitimate. 5. Restriction Definition

Identify materially restricted uses. 6. Exclusion Definition

Identify contexts, conditions, objects, scenarios, or claims outside the evidence basis. 7. Unsupported Interpretation Assessment

Identify interpretations that exceed the governed meaning. 8. Prohibited Use Definition

Define uses that cannot legitimately proceed on the existing simulation evidence basis. 9. Persistence Control

Ensure material limitations remain linked to evidence through communication, aggregation, transformation, and transfer. 10. Limitation State and Review

Assign a governed limitation state and identify conditions requiring reassessment.

Governance Outputs

SUM may produce: Simulation Intended Use Record Simulation Permitted Use Condition Simulation Evidence Restriction Record Simulation Evidence Exclusion Record Unsupported Interpretation Record Prohibited Simulation Evidence Use Record Simulation Extrapolation Limitation Simulation Decision-Class Limitation Simulation Configuration Limitation Simulation Scenario Limitation Simulation Environmental Limitation Simulation Temporal Limitation Simulation Uncertainty Limitation Simulation Evidence Dependency Limitation Simulation Limitation Persistence Record Simulation Use Limitation State Governed Simulation Use Limitation Record

These outputs provide the limitation basis required by:

SVIM — Simulation Validation Interface Module.

Enterprise Application

SUM enables organizations to solve a critical problem that occurs after simulation evidence has already been produced correctly:

the evidence may be governed correctly at its source but used incorrectly downstream.

A simulation team may clearly understand that a result applies only to: one configuration; one scenario family; one environment; one parameter range; one model version; one uncertainty range.

But after the result enters a dashboard, management report, validation package, regulatory submission, engineering decision, or automated evidence pipeline, those qualifications may disappear.

The result may remain numerically identical while its organizational meaning becomes distorted.

SUM prevents that separation.

For a spacecraft simulation, evidence concerning a defined landing envelope must not silently become evidence for all landing conditions.

For an autonomous vehicle, millions of successful simulated scenarios must not be represented as proving unrestricted real-world safety where the simulation did not represent all materially relevant conditions.

For a robot, evidence produced for one sensor, actuator, payload, software, and environmental configuration must not automatically transfer to another configuration.

For engineering simulation, results generated within defined physical assumptions and parameter ranges must not be presented as universal physical behavior.

SUM therefore ensures that organizations preserve not only:

the simulation result

but also:

the rules governing what that result may legitimately be used to mean.

Foundational Principles

A material use limitation is part of the evidential meaning.

Reliance boundaries must become operational use controls.

Evidence must not travel without the limitations that govern it.

Failure not observed does not mean failure impossible.

Simulation success does not independently establish operational safety.

Scenario success does not establish universal performance.

Comparative evidence does not automatically establish absolute performance.

Simulation prediction does not guarantee outcome.

Numerical precision does not create evidential certainty.

Extrapolation must remain identifiable as extrapolation.

Multiple dependent evidence objects must not be represented as independent confirmation.

Summarization must not remove limitations necessary for legitimate interpretation.

A limitation must travel at least as far as the evidence it limits.

Simulation-derived evidence must never become more general, certain, transferable, or authoritative merely because the limitations that originally constrained it have been forgotten.

Canonical Closing Statement

Simulation Use Limitation Module establishes the explicit evidence-use control layer within the Simulation Evidence & Reliance Boundary Standard by transforming governed reliance boundaries into persistent conditions, restrictions, exclusions, caveats, unsupported interpretations, prohibited uses, and review requirements attached to simulation-derived evidence. Through governance of intended use, permitted-use conditions, configuration, scenario, environmental, parameter, temporal, uncertainty, dependency, extrapolation, decision-class, communication, aggregation, and transfer limitations, SUM prevents simulation evidence from acquiring broader meaning merely through reuse, summarization, organizational transfer, automation, or loss of context. By distinguishing reliance boundaries from use limitations, output limitations from evidential limitations, and simulation-specific restrictions from broader validation determinations, SUM preserves the architectural boundaries of SIMULOS®, VALIDOS®, ORA™, INTEGROS®, ADIT®, AGA™, AIG®, and ASGA™. A simulation limitation is therefore not peripheral commentary: where it materially constrains evidential meaning, it is part of the governed evidence itself and must remain attached for as long and as far as that evidence is used.

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